Silicon Sampling on CES 2025: Three Ways to Extend the Model

New questions, longer context, and a dossier-based analyst · results so far

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1. What we tested

The test: predict how 15 subgroups (5 regions × 3 age groups) answered CES 2025 questions, using only data from before 2025. The score is **TV**, the share of answers placed in the wrong spot, averaged over subgroups and questions — lower is better. As a point of reference, we also compute the **floor**: how much a second real CES 2025 poll of the same size would miss by, just from ordinary sampling luck. No model can beat the floor; it is not a competitor, it is the ceiling of what is achievable.

We now score two sets of questions. The **repeated set** (10 questions) were all asked in CES 2021 too, so a model could, in principle, get credit for having seen them before. The **new set** (8 questions) were never asked in any of our 8 training surveys — US relations with the US since Trump's election; who is to blame for worse relations; protecting weaker nations from foreign aggression; whether safe injection sites make neighbourhoods unsafe; whether campus protests have gone too far; returning land and resources to Indigenous peoples; worry about one's own housing security; and whether government should guarantee decent housing for all. These are the closest thing we have to what a real client would ask: a question nobody has polled before.

Short label	Question	Answers
US relations	Has the relationship between Canada and the US become better or worse after Trump's election?	Better / worse / same
Who is to blame	Who is responsible for worsening relations between the US and Canada?	Canada / US / both
Protect weaker nations	Protecting weaker nations from foreign aggression, as a foreign policy goal	Very / somewhat / not important
Safe injection sites	Safe injection sites make neighbourhoods unsafe.	Agree, 5 points
Campus protests	Student protests on university campuses have gone too far.	Agree, 5 points
Indigenous land	Return control over land and natural resources to Indigenous peoples.	Support, 4 points
Housing worry	How worried are you about the security of your housing situation in the next few years?	Worry, 4 points
Housing for all	The government has a responsibility to provide decent housing for all.	Agree, 5 points

One caveat: "Who is to blame" was asked to only part of the sample, probably those who said relations got worse. The model answers it for everyone, which may unfairly inflate its error on that question.

We are testing three different ways of getting a model to produce these answers:

Approach	What the model sees	What it outputs
A. Role-play, short context	7 of one real person's own past answers	That one person's guessed answer
B. Role-play, long context	About 16 of one real person's own past answers	That one person's guessed answer
C. Dossier-to-distribution	A write-up of how Canadians and each subgroup answered similar past questions	The full percentage table for all 15 subgroups, in one go

A and B repeat this one-person-at-a-time guess 15,000 times (once per fake respondent) and tally the results into percentages. C skips the role-play: the model acts as an analyst and writes the table directly. For repeated questions, any past version of the question worded exactly the same is removed before the model sees anything, so even "repeated" questions are new to it.

A and B are trained the same way, on 50,000 to 150,000 answers from 8 Canadian surveys (2019–2024); each size is trained twice, because two identical training runs never give exactly the same model. C is tried three ways, always starting from the same Llama model as A and B. *Untrained*: the model reads the dossier as is. *Trained by example*: each of about 650 past survey questions becomes an exercise (its dossier, built without it, and the real table as the answer to copy). *Trained by reward*: the model writes 8 tables per exercise, a small program scores each one against the real table, and the model is trained only on its own best attempts; it never sees the real table.

2. Results

	Repeated (10)	New (8)	Same error as a real poll of… per subgroup (repeated)
A. Role-play, 7 answers	0.178	0.243	14
B. Role-play, 16 answers (100k)	0.178	0.242	14
C. Dossier, untrained Llama	0.196	0.264	11
C. Dossier, trained by example	**0.152**	**0.214**	**19**
C. Dossier, trained by reward	0.202	0.216	11
C. Dossier, untrained GPT-5.5 (flagged)	0.106	0.144	see note below
CES 2021 reused	0.151	n/a	20
National mode, for everyone	0.540	0.544	—
Floor (two halves of CES 2025)	0.062	0.074	—

A and B rows: 50,000 and 100,000 training examples. Every trained row except A is the mean of two training runs. B by training size:

	Repeated (10)	New (8)	New, without the 2 Trump questions
A. 7 answers, 50,000 examples	0.178	0.243	0.175
B. 16 answers, 50,000	0.212	0.263	0.194
B. 16 answers, 100,000	0.178	0.242	0.163
B. 16 answers, 150,000	**0.171**	0.267	0.177

Reading the last column of the first table: 0.151 is worth about 20 real respondents per subgroup (300 in total), 0.10 about 40 (600), 0.08 about 65 (1,000, a standard poll).

About the GPT-5.5 line: a recent model, shown muted as a caution, not a result: its data run into 2026 and may include 2025 opinion or CES 2025 itself (it is almost perfect on national defence spending).

New questions, one by one (B at 100,000 examples; B and C by example: mean of two runs):

New question	A. 7 answers	B. 16 answers	C. Dossier (untrained)	C. Dossier (by example)	Floor
US relations	0.416	0.529	0.445	0.345	0.030
Who is to blame	0.479	0.430	0.409	0.562	0.033
Protect weaker nations	0.172	0.131	0.331	0.069	0.045
Safe injection sites	0.173	0.173	0.240	0.147	0.104
Campus protests	0.220	0.192	0.222	0.112	0.102
Indigenous land	0.139	0.142	0.239	0.126	0.109
Housing worry	0.202	0.181	0.124	0.214	0.099
Housing for all	0.143	0.162	0.099	0.140	0.070
Mean	**0.243**	**0.242**	**0.264**	**0.214**	**0.074**

The two US–Trump questions are the clear outliers for every approach (0.41 to 0.57), and they move a lot from one training run to the next. On the other six new questions, A averages 0.175 — the same as on repeated questions — B at 100,000 examples 0.163, and C trained by example **0.135**.

3. What we learn so far

- **New questions are not the problem, 2025 events are.** The role-play model scores about the same on new and repeated questions, except for the two questions about US relations under Trump. Those ask about something that happened in 2025, after the model's training data ends — it cannot know the answer, no matter how the question is asked. Questions that are merely *new* in wording, but not about events after the model's knowledge cutoff, are no harder than ones it has seen before.
- **The trained analyst is our best approach.** C trained by example, on only about 650 past questions, matches CES 2021 reused on repeated questions (0.152 vs 0.151) without ever seeing them, and beats A and B on new ones (0.214 vs 0.24; 0.135 vs 0.16–0.18 without the Trump questions). Its two training runs agree closely (0.151 and 0.154), so this is not one lucky run.
- **Learning from its own best attempts helps less.** C trained by reward gains on new questions (0.264 → 0.216) but not on repeated ones (0.202, no better than untrained). Showing the model the real answer works better than letting it pick among its own guesses.
- **A model's own knowledge cutoff limits how it can be judged.** Models whose training data reach into 2025 or 2026 may already have absorbed 2025 opinion data, or the survey itself. A good score from such a model is not trustworthy evidence of genuine prediction, which is why we flag the GPT-5.5 result rather than count it as a result.
- **More about each person helps only with more training.** With 16 answers per person instead of 7, the model does worse at 50,000 examples (0.212 vs 0.178), as well at 100,000, and better at 150,000 (0.168, our best trained result on repeated questions, still behind CES 2021 reused at 0.151). On new questions the gain is small (0.163 vs 0.175 without the Trump questions).

4. Next steps

- Push C by example further: more exercises (more surveys, more variants per question) and the per-subgroup sample sizes in the dossier, and test it on a third set of questions.
- **Give the model the news it cannot know.** Add a short briefing on current events (tariffs, annexation threats, new prime minister), written only from facts published before the survey, and check that it fixes the two Trump questions without hurting the others.
- Put the effort into the dossier approach (C) rather than longer role-play context (B): C is better, much cheaper to run (one request per question instead of 15,000) and gives directly the table a client wants.